number of intersections in a candidate solution is more expensive with curve-line drawings.

In the next chapters, we will focus on straight-line drawings, but we'll also show how to extend them to curve-line graphs.

15.4 Edge Intersections

So, the last bit of code we need to make our initial fully-randomized algorithm for minimal-crossing embeddings is the method that checks if two edges intersect.

In this section we'll start discussing the variant for straight-line drawings: since edges are drawn as segments, it's easier to check if they intersect.

Then we'll move to Bézier curves, briefly explaining how they work, the subset of curves we'll allow, and how to check for intersections for elements of this subset.

15.4.1 Straight-line Segments

Let's start easy with the intersection between two segments. After all, we also have said a few times that we decided to focus on straight-line drawings, so we'll devote more space to discussing this case.

An initial strategy to cheaply screen pairs not intersecting is shown in figure 15.16: if we draw a box, parallel to the axes, around each of the two segments, then obviously the two segments can't cross if the boxes do not intersect; in turn, the two boxes can't intersect if the projections of the two